



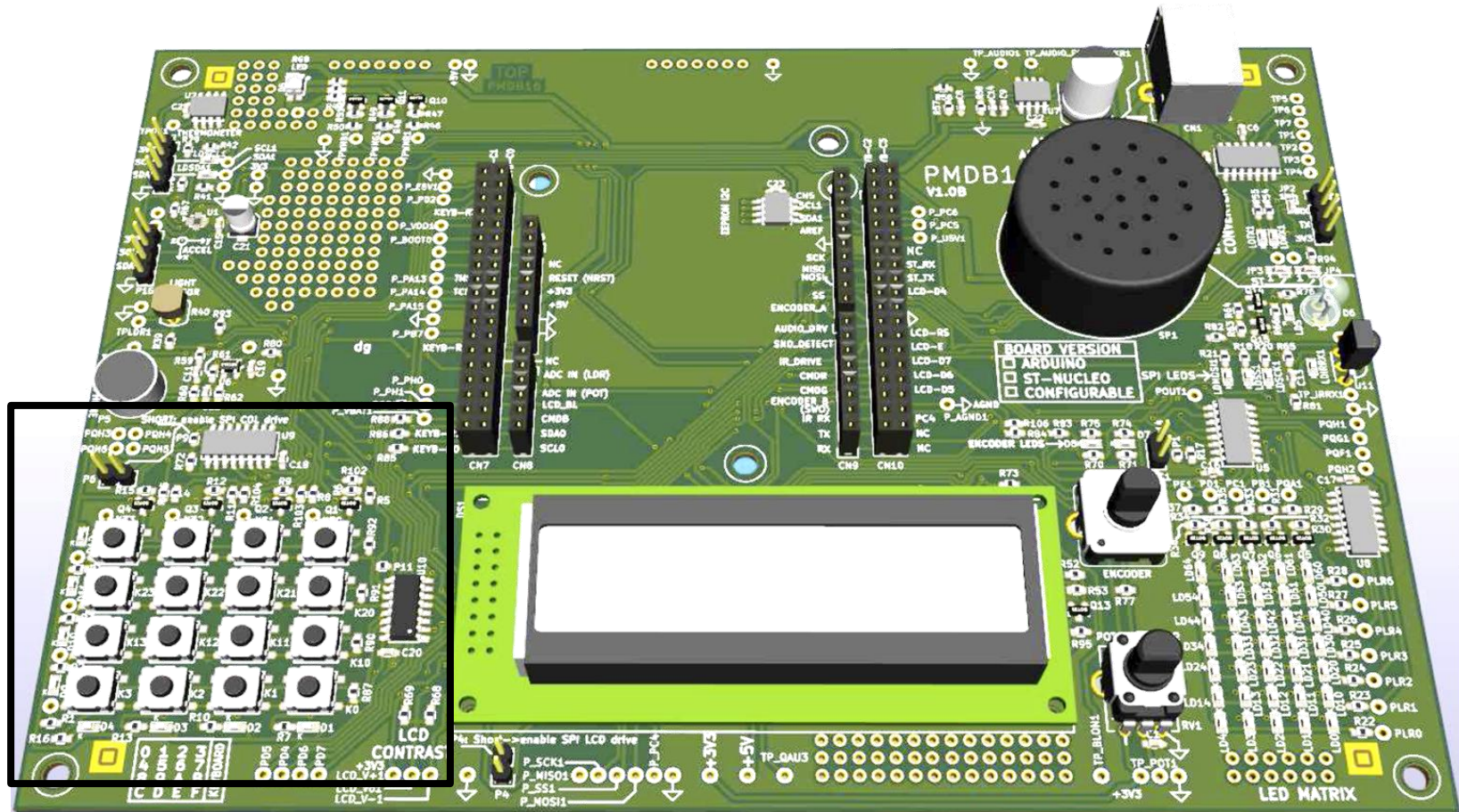
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STM32 – KEYBOARD

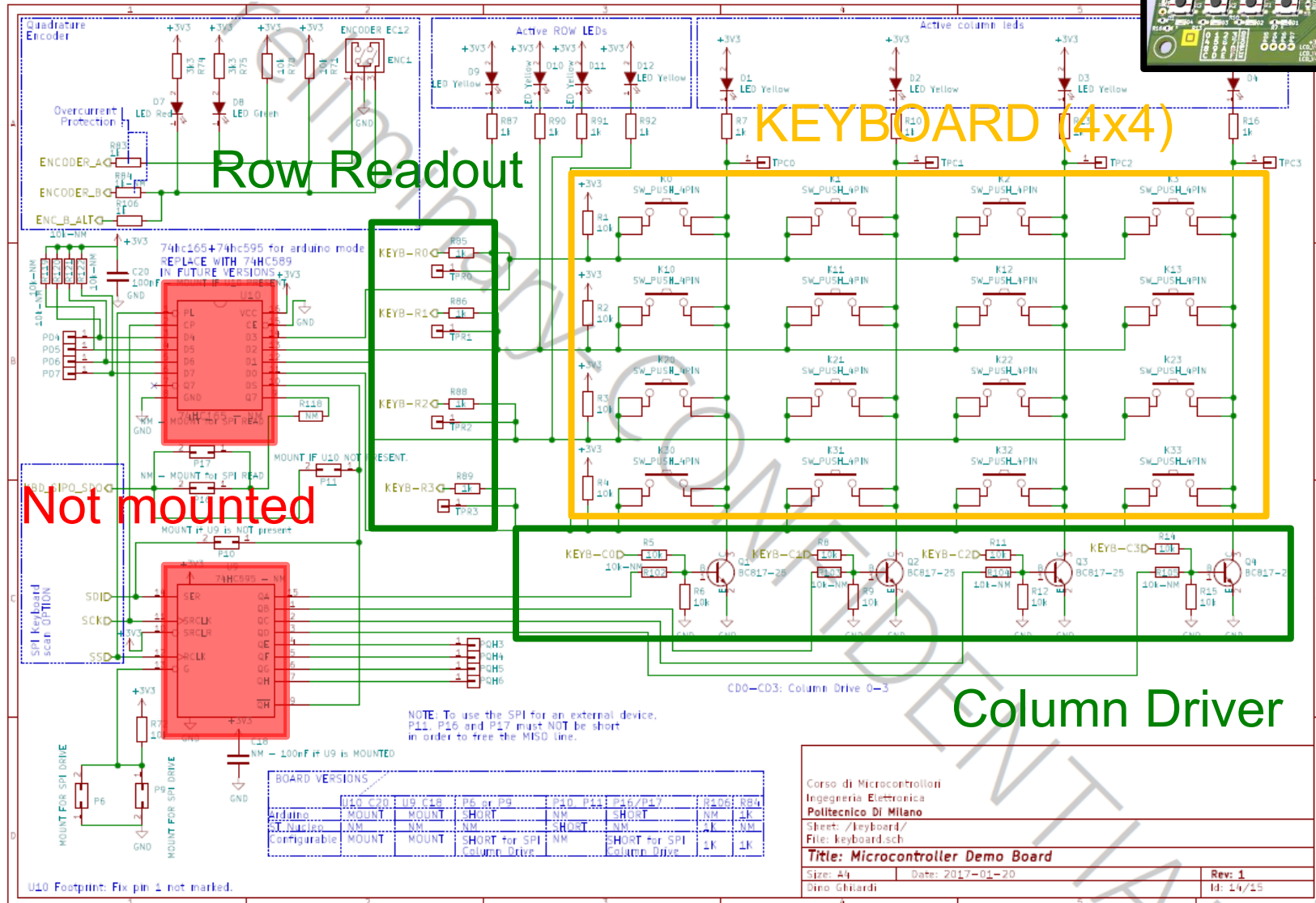
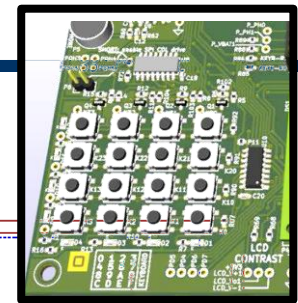
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KEYBOARD

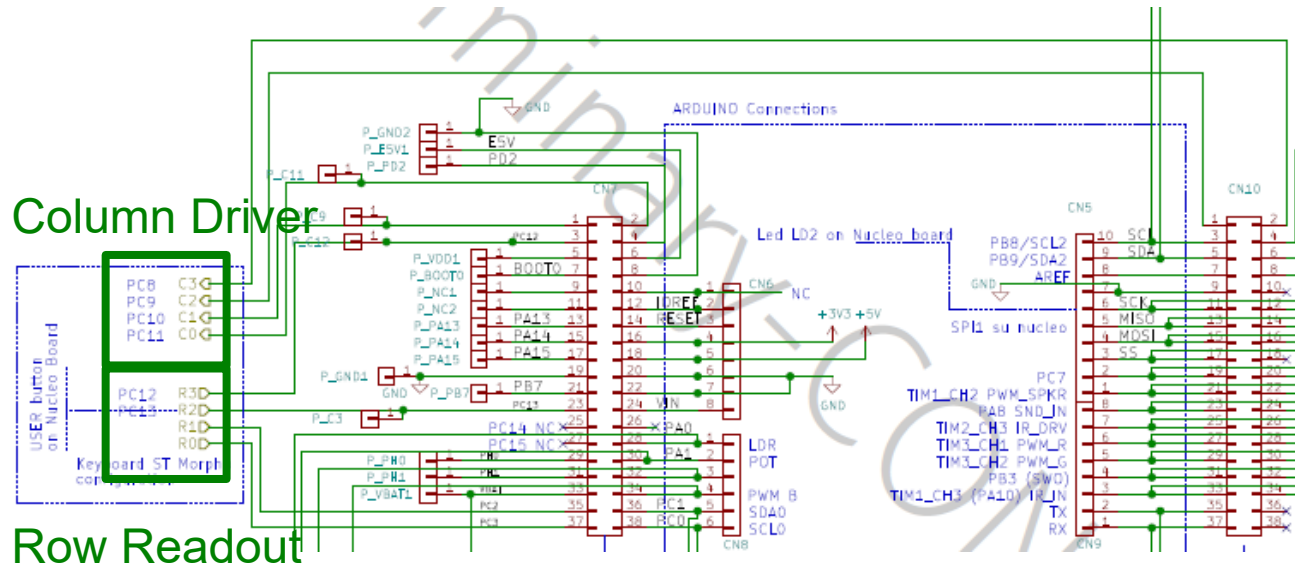


KEYBOARD – Schematic overview

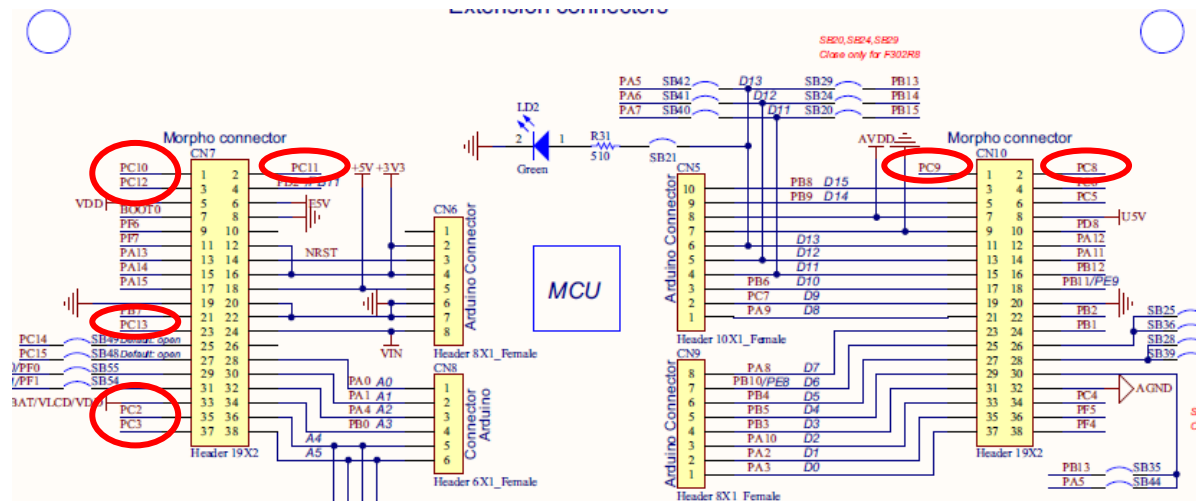




KEYBOARD – Pinout configuration



- PC2
 - PC3
 - PC12
 - PC13
- GPIO input
- PC8
 - PC9
 - PC10
 - PC11
- GPIO output





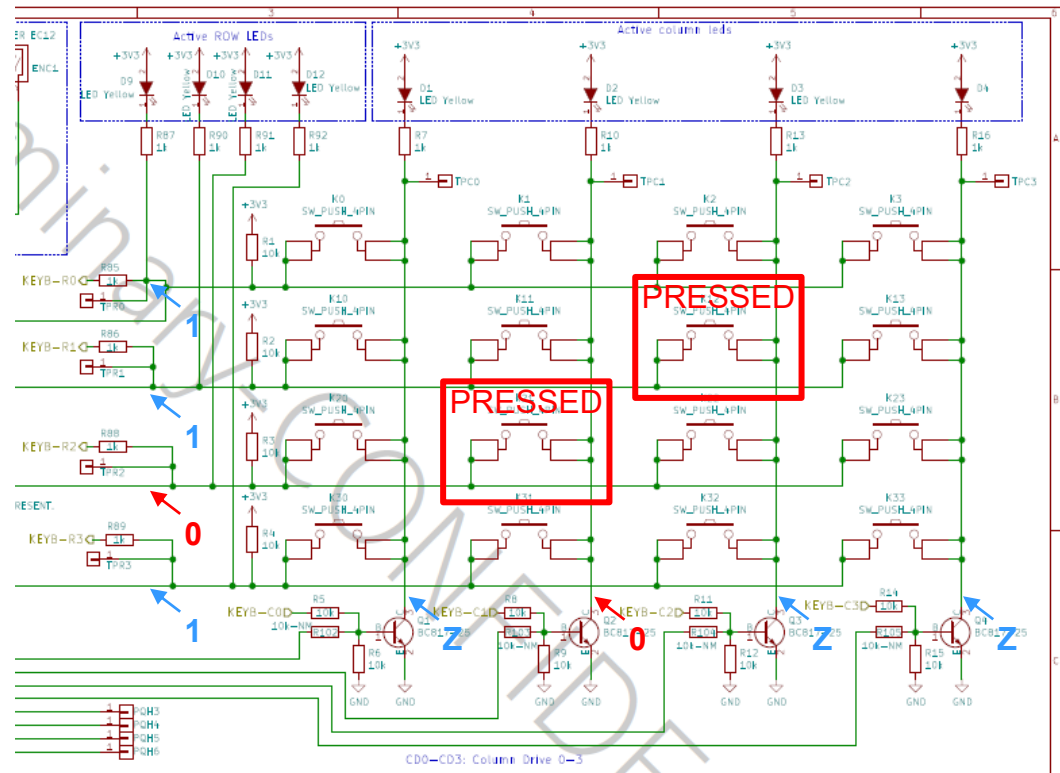
KEYBOARD – How to read a key

Pull-up resistors on the row lines

Open collector transistors on the column line pull it to ground or leave it floating

To read a key:

- **Scan** periodically across the keyboard by enabling **one column at a time**
- **Read the 4 rows:**
 - if a row is high (=‘1’) then the key is **NOT** pressed
 - Else, the key **IS** pressed
- To **debounce**, wait for a minimum keypress duration before validating the input





Project 1 – Keyboard readout

Objective

**Scan each column and read
the keyboard
using a timer interrupt**



Project hints

- Configure the 8 GPIOs required to scan the keyboard.
- Setup a timer to provide an interrupt every few ms. In the callback, acquire the status of the 4 rows and then enable a new column. Figure out a way to keep track of when each of the 16 keys is pressed and for how long.
- Within the loop(), if a key is pressed long enough, send the corresponding character to PC via UART. Do not repeat the key send if the key remain pressed. Send the code again if the key is released and pressed again.
- Compile and debug the code. (Suggestion: step through each column slowly and check the code operation)



Project 1 - Setting on CUBE pinout (1)

Starting from a default model project

Select the following
GPIO_input and
GPIO_output ports

